



What is a gauge theory?

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- Field theory

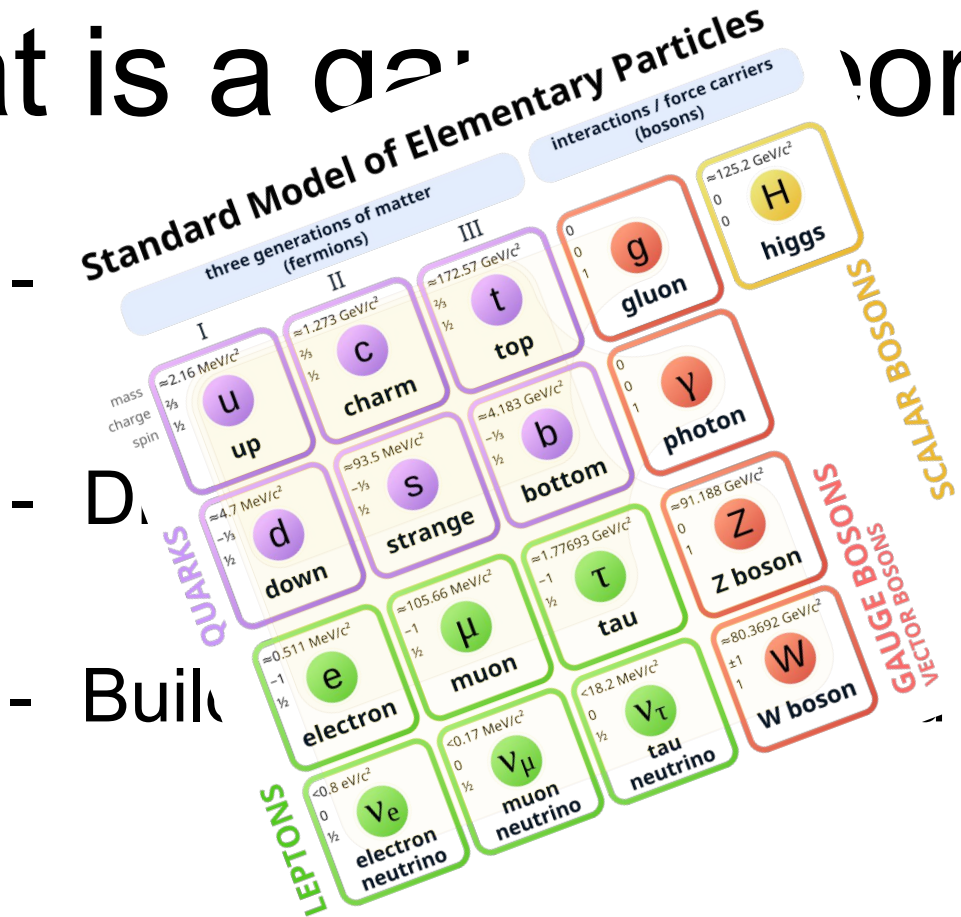
What is a gauge theory?

- Field theory
- Discovered in the 60s

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- Field theory
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- Building block of Standard Model

What is a $\alpha\alpha$ theory?

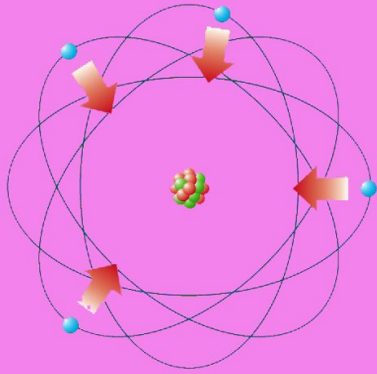


- D

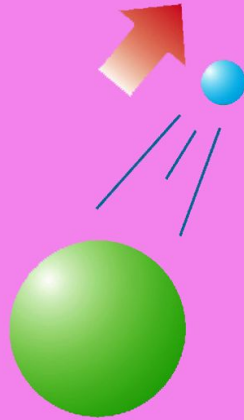
- Built

- model

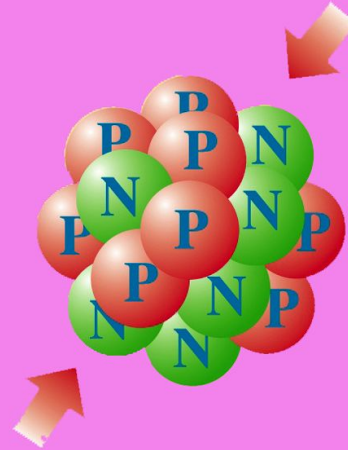
FUNDAMENTAL FORCES OF NATURE



**Electro-
magnetism**



**Weak
Interaction**

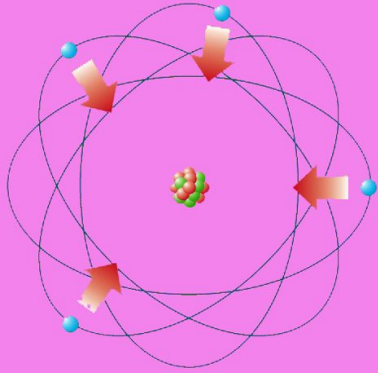


**Strong
Interaction**

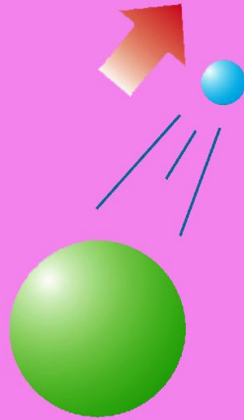


Gravitation

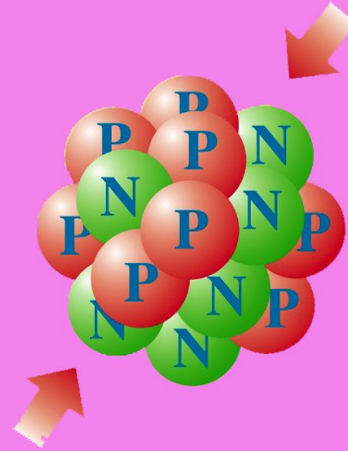
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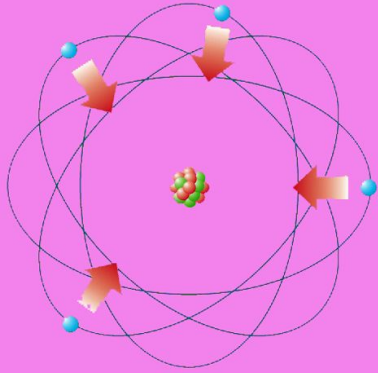


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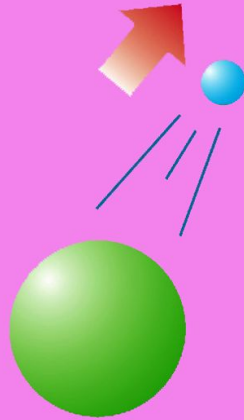


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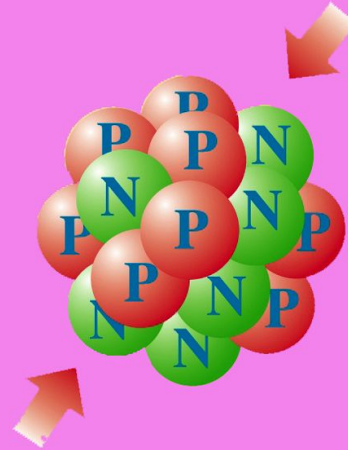
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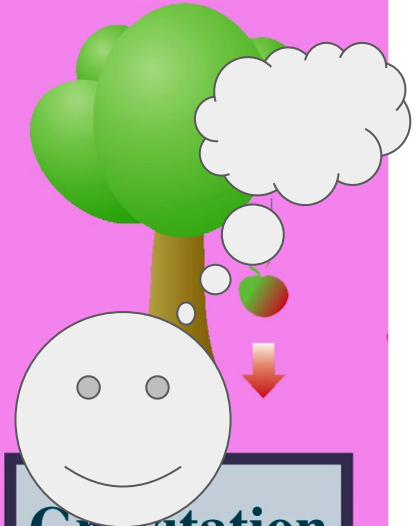
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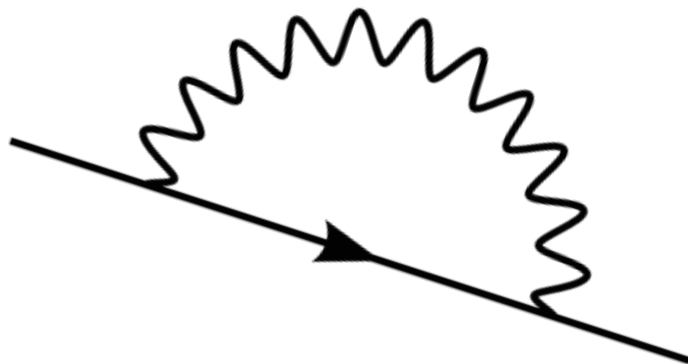
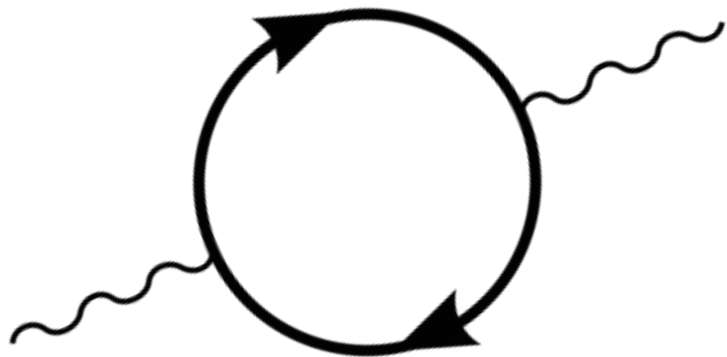
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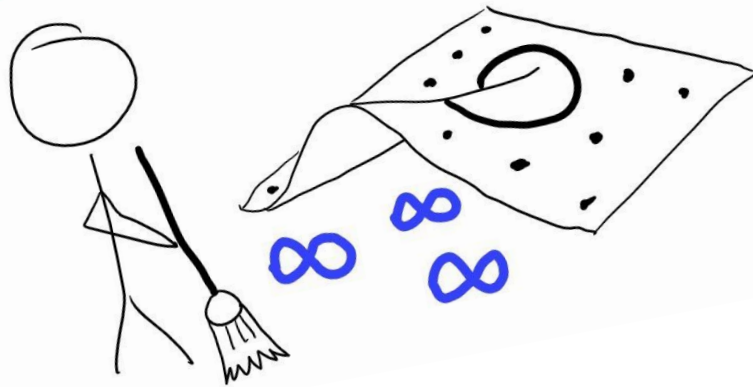
**Strong
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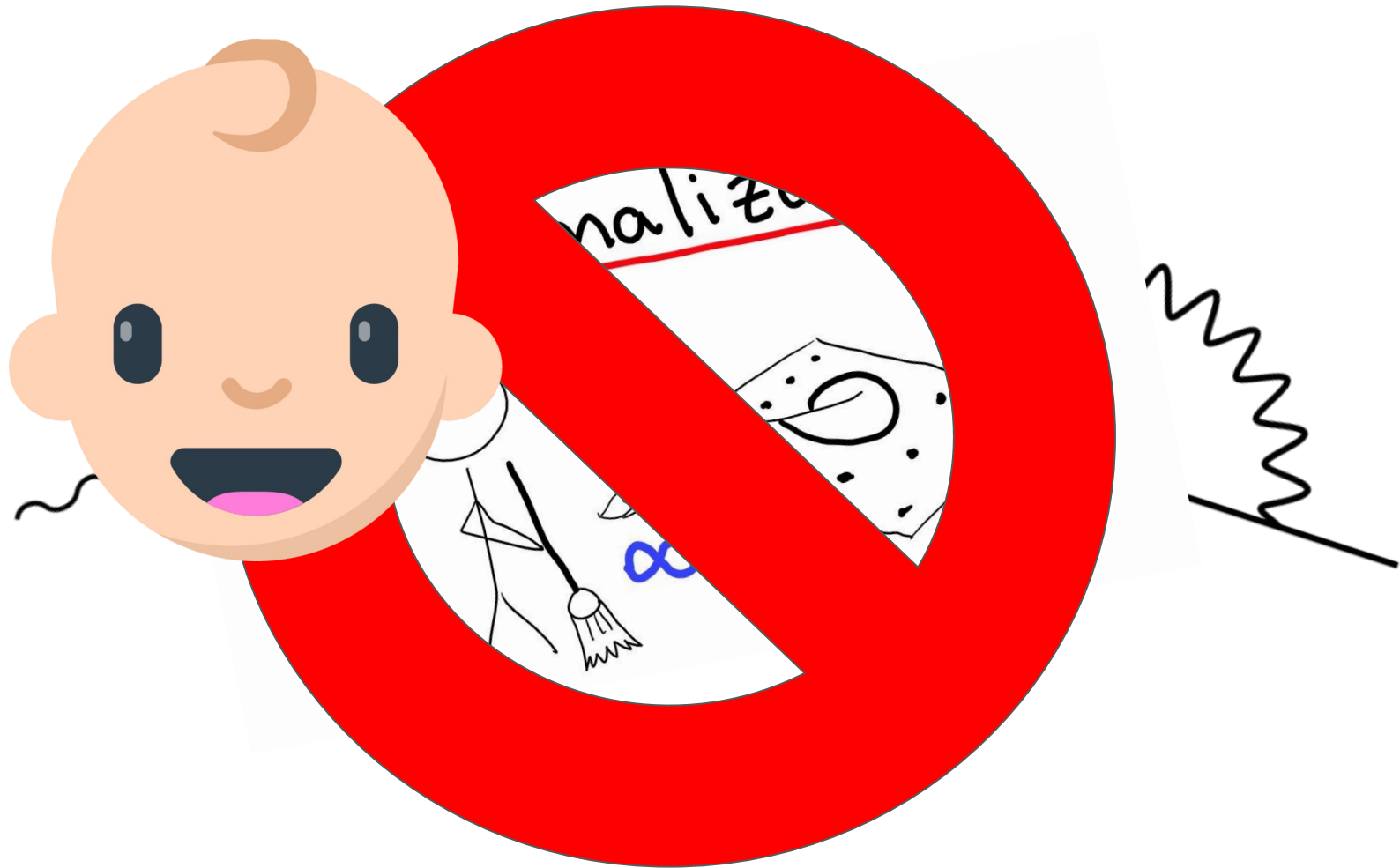


Gravitation



Renormalization





Symmetry group	$SO(1,3) \subset GL(1,3)$	$U(1)$	$SU(2)$
Covariant derivative			
Tensorial	$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu_{\mu\lambda} V^\lambda$	$D_\mu \psi = \partial_\mu \psi + A_\mu \psi$	$D_\mu \psi = \partial_\mu \psi + A^a_\mu T^a \psi$
Cartan	$\nabla V = dV + \omega \wedge V$	$D\psi = d\psi + A\psi$	$D\psi = d\psi + A \wedge \psi$
1-form			
Tensorial	$\Gamma^\lambda_{\mu\nu}$	A_μ	A^a_μ
Cartan	$\omega^a_b = \Gamma^a_{b\mu} \theta^\mu$	$A = A_\mu \theta^\mu$	$A^a = A^a_\mu \theta^\mu$
2-form			
Tensorial	$R^\lambda_{\mu\rho\nu} = \partial_\mu \Gamma^\lambda_{\nu\rho} - \partial_\nu \Gamma^\lambda_{\mu\rho} + \Gamma^\lambda_{\mu\sigma} \Gamma^\sigma_{\nu\rho} - \Gamma^\lambda_{\nu\sigma} \Gamma^\sigma_{\mu\rho}$	$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$	$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + f^{abc} A^b_\mu A^c_\nu$
Cartan	$\Omega^a_b = \frac{1}{2} R^a_{b\mu\nu} \theta^\mu \wedge \theta^\nu$	$F = \frac{1}{2} F_{\mu\nu} \theta^\mu \wedge \theta^\nu$	$F^a = \frac{1}{2} F^a_{\mu\nu} \theta^\mu \wedge \theta^\nu$
	$\Omega^a_b = d\omega^a_b + \omega^a_c \wedge \omega^c_b$	$F = dA + A \wedge A$	$F^a = dA^a + A^a \wedge A^a$

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Cartan	$\Omega^a_b = \frac{1}{2} R^a_{b\mu\nu} \theta^\mu \wedge \theta^\nu$ $\Omega^a_b = d\omega^a_b + \omega^a_c \wedge \omega^c_b$	$F = \frac{1}{2} F_{\mu\nu} \theta^\mu \wedge \theta^\nu$ $F = dA + A \wedge A$	$F^a = \frac{1}{2} F^a_{\mu\nu} \theta^\mu \wedge \theta^\nu$ $F^a = dA^a + A^a \wedge A^a$